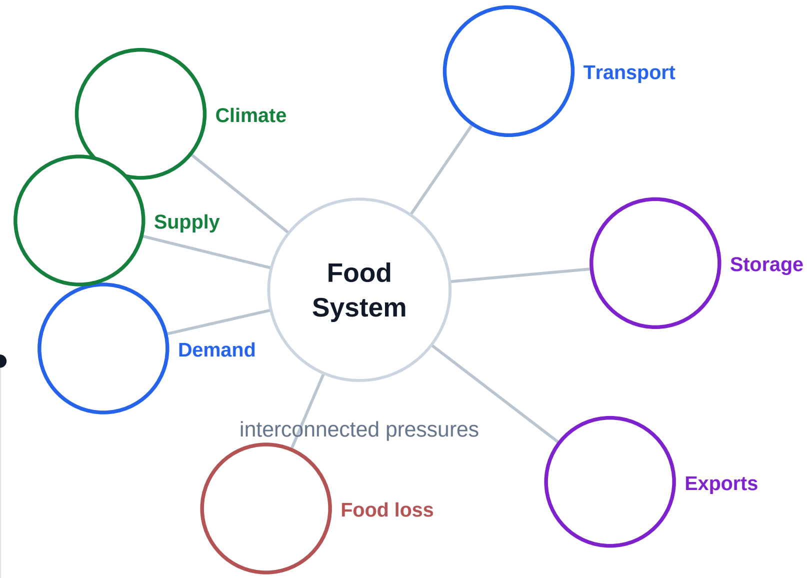
ALGEBRA 2 CUMULATIVE SEMESTER FINAL PROJECT

Systems Under Pressure

The Mathematics of Food Systems



> **"All models are wrong, but some are useful."**
>
> George Box

A spacious mathematical field notebook for sketching, modeling, revising, and explaining.

JOURNAL NORMS

Rough work belongs here.

This journal is not a worksheet packet. It is the visible record of your group's evolving mathematical thinking.

Value uncertainty

A strong investigation often begins confused. Label assumptions, questions, disagreements, and places where a model stops making sense.

Revise visibly

Cross out, annotate, draw arrows, compare versions, and explain why your thinking changed.

Models are arguments

A model is not an answer. It is a claim built from assumptions, evidence, representations, and limits.

Field note style

Messy is acceptable. Hidden thinking is not. The strongest journals show graph, algebra, context, units, limitations, and revision together.

USE MODEL

Shared field notebook. Individual accountability.

The project is collaborative, but grading is individual. The journal and individual evidence serve different purposes.

Physical group journal

The physical journal captures the group's evolving mathematical thinking. It should look used: sketched on, annotated, revised, taped into, crossed out, and questioned.

Individual digital evidence

Your individual digital evidence captures your own mathematical reasoning inside the group investigation. This may include personal graph notes, recalculations, interpretations, AI critiques, limitation reflections, and explanations of how your thinking changed.

How they work together

The group journal preserves the messy investigation trail. Individual evidence shows how each student reasoned, interpreted, revised, and contributed mathematically inside that shared work.

Field notebook rule

Clean pages are not the goal. Crossed-out thinking belongs here. Strong investigations often become more nuanced, not more certain.

RUBRIC ALIGNMENT

Use the journal to make standards visible.

The rubric does not only score final answers. It looks for reasoning, interpretation, revision, limitations, precision, and flexible thinking across the investigation.

1. Sequences and Series

Content Standard

2. Exponentials and Logarithms

Content Standard

3. Polynomials

Content Standard

4. Mathematical Communication and Interpretation

Skill Grade

5. Comparing Models, Assumptions, and Limitations

Skill Grade

6. Algebraic Precision and Flexible Number Sense

Skill Grade

CLASS 1

Entering the System

What relationships are we beginning to notice?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

SYSTEM MAP CANVAS

Sketch variables, links, pressures, unknowns, and possible feedback loops.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Variable | Units | Why it matters | What is hard to measure. Which variables seem politically, socially, or physically difficult to measure?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

SKETCH BEFORE TECHNOLOGY

Predict the shape of one relationship before using Desmos, Sheets, or another tool. What relationship seems obvious at first but may be misleading?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Discuss uncertainty, early assumptions, possible misleading patterns, and variables that may be hard to measure reliably.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

CLASS 2

Modeling Change

How is the system changing over time?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

Does your context change in steps, continuously, or both? Justify the representation that best fits your system.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Does the system behave additively, multiplicatively, unpredictably, or in phases? What evidence supports this?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Write useful formulas if the pattern supports them. Annotate what each term, parameter, and assumption means in your context.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

What does accumulated change represent in your system? Would accumulation realistically continue forever? Why or why not?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

MEANINGFUL TIME SCALES

Choose multiple time ranges that matter for your investigation. How does the model behave differently across those scales?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

PATTERN BREAKDOWN

When does the pattern stop behaving consistently? What evidence would make you revise or reject it?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

CLASS 2 MODEL COMPARISON STUDIO

Changing assumptions creates different mathematical realities.

Choose model versions that are meaningful for your context. Which model is more useful, more realistic, or more honest about uncertainty? Which model hides constraints, exaggerates certainty, or breaks first?

version A

version B

version C

CLASS 3

Acceleration, Decay, and Thresholds

When does repeated change become accelerated or compounding?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

What evidence suggests the change depends on the current amount rather than a constant added amount?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

SAMPLED VALUES VS CONTINUOUS MODEL

How does continuous exponential modeling change the interpretation compared to sampled or discrete terms in your context?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

BUILD THE MODEL

Define starting value, growth or decay factor, variables, assumptions, and realistic domain. Which assumption is doing the most work?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Use logarithms when you need to solve for an unknown time or threshold. What does that threshold mean in the real system?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

RATE OR ASSUMPTION COMPARISON

Compare multiple realistic rates or assumptions relevant to your investigation. Which assumptions cause the prediction to change dramatically?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Where does exponential behavior stop being realistic? What real-world forces might interrupt it? What hidden variables matter?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

CLASS 3 MODEL COMPARISON STUDIO

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Choose model versions that are meaningful for your context. Which model is more useful, more realistic, or more honest about uncertainty? Which model hides constraints, exaggerates certainty, or breaks first?

version A

version B

version C

CLASS 4

Constraints, Capacity, and Cost

How do physical and economic constraints reshape the system?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

FLAT BLUEPRINT

Original dimensions, square cutout variable, folding lines, units in centimeters.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

FOLDED STORAGE STRUCTURE

Height, interior dimensions, usable space, loading floor.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Factored form, standard form, and why multiplying dimensions creates a cubic.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

GRAPH ANNOTATION

Intercepts, multiplicity, extrema, domain, unrealistic regions, physical meaning.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Degree, even/odd, leading coefficient, as x $\to \infty$ and as x $\to -\infty$.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

How could division help analyze storage, packing, capacity, grouping, or remaining space? Interpret the quotient and remainder physically.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

CONSTRAINT REVISION

Change a realistic physical or economic constraint from your context. What tradeoff appears, and which graph regions become impossible?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Changing assumptions creates different mathematical realities.

Choose model versions that are meaningful for your context. Which model is more useful, more realistic, or more honest about uncertainty? Which model hides constraints, exaggerates certainty, or breaks first?

version A

version B

version C

CLASS 5

Revision, Synthesis, and Communication

What story does our mathematics tell?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

SYNTHESIS TABLE

Lens | What did it reveal? | Where did it fail?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Compare mathematical lenses and assumptions.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

FINAL MATHEMATICAL CLAIM

Use evidence, assumptions, limitations, tradeoffs, and implications. What remained uncertain even after modeling?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Choose a format. Plan how mathematical evidence, revisions, and uncertainty will be visible.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

IF WE REPEATED THIS INVESTIGATION

With better data, what would change? Which assumption, measurement, or model would you revisit first?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

INDIVIDUAL ARTIFACT SNAPSHOT

Revision, hidden-variable analysis, limitation critique, or r/R^2 interpretation. A high R^2 can still mislead; correlation can support a claim without proving causation.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?